

Harika Uppu

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Professional Summary

Embedded Firmware Engineer with experience in STM32 and PIC microcontrollers based on ARM Cortex architecture. Skilled in Embedded C programming, peripheral driver development, and interrupt-driven firmware design. Experienced in SPI/UART communication, RTOS-based task management, and low-level debugging. Passionate about building reliable real-time embedded systems and optimizing firmware performance.

Work Experience

Tech Mahindra

Associate Software Engineer | Hyderabad

Feb 2024 – Present

- Developed SPI-based communication drivers for master-slave embedded systems enabling reliable full-duplex data transfer.
- Configured SPI peripheral parameters including clock polarity (CPOL), phase (CPHA), chip select, and baud rate.
- Implemented SPI communication with external EEPROM (25AA160A) for non-volatile data storage, including read/write operations and command handling (WREN, READ, WRITE).
- Developed low-level SPI driver routines using register-level programming for data transmission and reception.
- Integrated SPI driver with RTOS tasks and used synchronization mechanisms like semaphores for safe data access.
- Debugged SPI communication issues using logic analyzer/oscilloscope by analyzing MOSI, MISO, SCLK, and CS signals.
- Worked with RTOS-based device driver interfaces in systems using QNX.

Projects

Embedded Projects

ARM Cortex-M4 (STM32F407) – Bare-Metal Firmware Development

- Developed Embedded C firmware using direct register-level programming.
- Configured GPIO, UART, SPI, I2C, Timers, ADC, and PWM peripherals.
- Implemented interrupt-driven execution using NVIC and peripheral interrupts.
- Developed UART-based serial communication for real-time debugging.
- Structured firmware into modular driver and application layers
- Performed on-board validation using JTAG and serial debugging tools.

PIC16F877A – Embedded C Firmware Implementation

- Developed Embedded C applications using MPLAB XC8 compiler.
- Configured ADC module for analog signal acquisition and processing.
- Implemented PWM generation for output control applications.
- Designed interrupt-based execution flow using external and timer interrupts.
- Conducted simulation-based testing and functional validation.

UART Communication Implementation – STM32

- Configured UART peripheral for full-duplex serial communication.
- Implemented interrupt-based data transmission and reception.
- Validated communication through serial terminal output.
- Debugged baud rate and framing configurations for reliable communication.

Academic Project:

Telugu Automatic Speech Recognition System

Osmania University | Ubuntu | Kaldi

- Developed continuous speech-to-text model using Kaldi toolkit.
- Performed dataset preprocessing, segmentation, model validation, and tuning.
- Worked in Linux-based development environment.

Skills

- **Programming:** C Programming, Embedded C, Python
- **Embedded Concepts:** Bare Metal Programming, Device Drivers, Interrupt Handling
- **Microcontrollers:** ARM Cortex-M4 (STM32F407), PIC16F877A, 8051.
- **Peripherals:** GPIO, Timers, Interrupts, ADC, PWM, Clock Configuration, Memory-Mapped Registers
- **Communication Protocols:** UART, SPI, I2C,
- **Debugging & Development Tools:** Git, Jira, JTAG Debugging, Serial Debugging, MPLAB XC8, STM32CubeIDE, Keil, VS Code
- **Operating Systems:** Linux, Windows, RTOS, QNX

Internship

BSNL Intern

May 2022 – June 2022

BSNL | Eluru, Andhra Pradesh

- Gained knowledge of optical fiber communication and GSM modules.
- Assisted in installation, maintenance, and troubleshooting of fiber networks.

Automation, APSSDC

September 2022 – October 2022

- Worked on PLC and SCADA systems for industrial automation.

Embedded Systems Training & Certifications

Embedded Systems Training – Argyn Technologies

- PIC16F877A Microcontroller Programming
- ARM Cortex-M Programming
- Embedded C & Peripheral Interfacing

Embedded Hardware Foundation & Basics of Electronics & Microprocessor Internals – Etalvis

- Digital & Analog Electronics Fundamentals
- Circuit analysis and hardware understanding
- Microcontroller hardware interfacing concepts

Education

Bachelor of Technology (B. Tech, ECE)

September 2019 - July 2023

Adikavi Nannaya University, Rajahmundry (India) – CGPA: 7.6

Cloud & IT Certifications

- AZ-900
- Google Cloud Practitioner
- AWS Cloud Practitioner
- Python 101 for Data Science